Z-Chain: A Universal Zero-Knowledge Proof Platform with Post-Quantum Security on Zoo Network

Antje Worring & Zach Kelling

Zoo Labs Foundation

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Abstract

We present Z-Chain, the zero-knowledge proof substrate of Zoo Network, providing universal proving, cheap verification, and composable proofs across all Zoo chains. Z-Chain implements a dual-lane architecture: a *production lane* (Groth16, PLONK, STARK) for stable deployments, and a *research lane* for experimental proof systems with versioned capability identifiers. The core primitive is the *Universal Receipt*—a chain-native proof-of-proof object that enables recursive composition, cross-chain verification, and long-term auditability. We integrate post-quantum security through Poseidon2 hash commitments (replacing Pedersen, achieving 2000× speedup at 7.5× lower gas cost), ML-DSA signature verification within ZK circuits, and STARK proofs over the Goldilocks field for transparent, quantum-resistant proving. The PQZ (Post-Quantum Zero-Knowledge) algorithm combines these primitives into a unified framework where: (1) proofs are generated using STARKs with Poseidon2 commitments, (2) verification uses ML-DSA-signed receipts, (3) cross-chain export uses Groth16 wrappers for EVM compatibility, and (4) private computation uses FHE-ZK hybrid protocols on Zoo's T-Chain. We benchmark Z-Chain at 12,000 proof verifications per second on GPU-accelerated validators with 1.8-second end-to-end latency from proof submission to receipt finalization. Formal verification in Lean 4 proves soundness, completeness, and zero-knowledge properties for all three production proof systems. This work establishes the first post-quantum zero-knowledge infrastructure integrated natively into a blockchain consensus layer.

1 Introduction

Zero-knowledge proofs enable a prover to convince a verifier of a statement's truth without revealing the underlying witness. In blockchain systems, ZKPs serve three critical roles: scaling (rollup compression), privacy (confidential transactions), and verifiable computation (off-chain execution with on-chain verification). However, existing ZK infrastructure suffers from three limitations:

1. **Quantum vulnerability.** Groth16 and PLONK rely on elliptic curve pairings (BN254, BLS12-381) that are broken by Shor's algorithm on a sufficiently large quantum computer. As NIST has standardized post-quantum algorithms (ML-KEM, ML-DSA, SLH-DSA), ZK systems must migrate to quantum-resistant primitives.

2. **Fragmentation.** Each proof system (Groth16, PLONK, STARK, Nova, Halo2) produces incompatible proof formats. Cross-system composition requires manual bridging. No standard receipt format exists for proof-of-proof attestation.

*research@zoo.ngo

3. **Centralized infrastructure.** Proof generation is computationally intensive, concentrating proving power in well-funded operators. Decentralized proving markets exist (=nil;, Gevulot) but lack native blockchain integration.

Z-Chain addresses all three through a chain-native ZK substrate that is post-quantum by default, universally composable via receipts, and decentralized through Zoo Network's Proof of AI validator set.

2 Architecture

2.1 Chain Placement

Z-Chain operates as a specialized chain within the Zoo Network multi-chain architecture:

- **C-Chain** (Contract Chain): Application deployment. Consumes Z-Chain proofs via precompiles.
- **Z-Chain** (ZK Substrate): Proof registration, verification, and receipt generation. This paper's focus.
- **T-Chain** (Threshold Chain): FHE decryption coordination for private ZK-FHE hybrid protocols.
- **K-Chain** (Key Chain): Post-quantum key management (ML-KEM, ML-DSA) for proof signing.

2.2 Two-Lane Design

2.2.1 Production Lane

Three proof systems are permanently supported with stable verifier contracts:

System	**Setup**	**PQ-Safe**	**Verify Gas**
Groth16	Trusted	No†	230K
PLONK	Universal	No†	350K
STARK	Transparent	Yes	500K

Table 1: Production proof systems. †: Wrapped in PQZ for quantum resistance.

2.2.2 Research Lane

Experimental proof systems register under versioned capability IDs (≥ 100). Each carries an explicit risk label (`EXPERIMENTAL`, `AUDITED`, `DEPRECATED`) and runs in sandboxed verifier contracts that cannot affect production lane state.

3 The PQZ Algorithm

Post-Quantum Zero-Knowledge (PQZ) combines three primitives into a unified framework:

3.1 Poseidon2 Commitments

We replace Pedersen commitments (which rely on discrete log hardness) with Poseidon2 [1], a ZK-friendly algebraic hash function secure under collision resistance assumptions that do not require elliptic curve hardness.

Definition 1 (Poseidon2 Commitment)**.** *For message* $m \in \mathbb{F}_p^n$ *and randomness* $r \in \mathbb{F}_p$*:*

$$Commit(m, r) = Poseidon2(m \| r)$$

Performance. Benchmarks on Zoo Network validators (NVIDIA H100):

Operation	Poseidon2	Pedersen	Speedup
Commitment	29 ns	92 μs	2000×
Note commit	50 ns	99 μs	1980×
Merkle path (depth 32)	928 ns	2.9 ms	3125×
Gas cost	800	6,000	7.5×

Table 2: Poseidon2 vs Pedersen on GPU-accelerated validators.

3.2 STARK Proofs over Goldilocks

For the core proving system, PQZ uses STARKs over the Goldilocks field ($p = 2^{64} - 2^{32} + 1$), which provides:

- **Transparency**: No trusted setup. Verification relies only on hash function collision resistance.
- **Post-quantum security**: Security reduces to hash collision resistance (Poseidon2), not discrete log or factoring.
- **Efficient FRI**: The Goldilocks field admits 2^{32}-th roots of unity, enabling efficient Fast Reed-Solomon Interactive Oracle Proofs.

3.3 ML-DSA Receipt Signing

After STARK verification, the Z-Chain validator set signs the resulting receipt using ML-DSA (FIPS 204) threshold signatures coordinated through K-Chain:

1. Prover submits STARK proof to Z-Chain.
2. Z-Chain validators independently verify the proof.
3. Upon $\geq 67\%$ agreement, validators produce a threshold ML-DSA signature on the receipt.
4. The signed receipt is stored on-chain and indexed by receipt hash.

3.4 Groth16 Export for External EVMs

For interoperability with Ethereum and other EVM chains that only support BN254 pairings, PQZ provides a Groth16 export:

1. Take a verified STARK receipt from Z-Chain.
2. Generate a Groth16 proof that "I know a STARK proof that was verified and signed by the Zoo validator set."

3. Export the Groth16 proof to the target EVM chain.

This wraps the post-quantum STARK in a classical Groth16 for compatibility, while the original STARK receipt remains the source of truth on Z-Chain.

4 Universal Receipt Format

The core interoperability object:

Definition 2 (Universal Receipt)**.** *A receipt R consists of:*

$$\begin{aligned} R = (&\texttt{programId},\ \texttt{claimHash},\ \texttt{receiptHash},\\ &\texttt{proofSystemId},\ \texttt{version},\ \texttt{verifiedAt},\\ &\texttt{parentReceipt},\ \texttt{aggregationRoot}) \end{aligned}$$

where:

- `programId`: *Hash of the verified program/circuit.*
- `claimHash`: *Hash of public inputs.*
- `receiptHash`: *Self-referential hash ($H(R \setminus \texttt{receiptHash})$).*
- `proofSystemId`: 1*=STARK,* 2*=Groth16,* 3*=PLONK,* 4*=Nova,* ≥ 100*=research.*
- `parentReceipt`: *For recursive composition.*
- `aggregationRoot`: *Merkle root for batch receipts.*

5 ZK Precompile ISA

Z-Chain exposes cryptographic primitives as EVM precompiles at address range `0x0500-0x051F`:

Address	**Operation**	**Gas**
`0x0501`	Poseidon2 hash	800
`0x0502`	Pedersen commit (legacy)	6,000
`0x0510`	STARK field arithmetic (Goldilocks)	8–15
`0x0511`	STARK extension field (Fp2)	25
`0x0512`	STARK Poseidon2 over Goldilocks	800
`0x0513`	STARK FRI fold	2,000
`0x0514`	STARK Merkle verification	400/level
`0x051F`	STARK full proof verify	100K–500K

Table 3: ZK Cryptographic ISA precompiles.

6 FHE-ZK Hybrid Protocols

For private computation, Z-Chain integrates with T-Chain's threshold FHE:

1. **Encrypted input**: User encrypts data under Zoo's public FHE key.
2. **Homomorphic computation**: T-Chain validators evaluate the circuit on encrypted data.

3. **ZK proof of correctness**: A STARK proof attests that the computation was performed correctly on the ciphertext.
4. **Threshold decryption**: T-Chain validators (67-of-100 quorum) decrypt only the result, not the input.
5. **Receipt**: Z-Chain issues a receipt linking the FHE output to the ZK proof.

This enables verifiable private inference: a model runs on encrypted patient data, the output is decrypted, and a ZK proof certifies correctness—all without exposing the data.

7 Formal Verification

All critical Z-Chain components are verified in Lean 4:

Theorem 1 (STARK Soundness). *For any polynomial commitment scheme* Π *over the Goldilocks field with security parameter* λ*, the probability of a computationally bounded adversary producing a valid STARK proof for a false statement is at most* $2^{-\lambda}$.

Theorem 2 (Receipt Integrity). *For any receipt* R *with valid ML-DSA threshold signature from* $\geq 67\%$ *of validators, the probability that* R.`claimHash` *does not correspond to a correctly verified proof is at most* 2^{-128}.

Theorem 3 (PQZ Composition). *If the inner STARK proof is sound and the outer Groth16 wrapper is sound, then the composed PQZ proof is sound. The zero-knowledge property is preserved through both layers.*

The full proofs are available at `zoo-formal/lean/ZK/` in the Zoo formal verification repository.

8 Empirical Evaluation

8.1 Throughput

On a cluster of 5 Zoo validators (each with NVIDIA H100 80GB):

Operation	**Throughput**	**Latency**	**Gas**
STARK verify (simple)	12,000/s	1.1s	100K
STARK verify (complex)	3,200/s	1.8s	500K
Groth16 verify	28,000/s	0.4s	230K
PLONK verify	18,000/s	0.6s	350K
Receipt generation	8,500/s	1.4s	50K
PQZ export (STARK→Groth16)	450/s	8.2s	800K

Table 4: Z-Chain verification throughput on GPU validators.

8.2 Comparison

9 Conclusion

Z-Chain provides the first post-quantum zero-knowledge infrastructure integrated natively into a blockchain consensus layer. The PQZ algorithm unifies STARK transparency with ML-DSA quantum resistance and Groth16 EVM compatibility. Universal receipts enable composable

	Z-Chain	**=nil;**	**Polygon**	**StarkNet**
PQ-safe	Yes	No	No	Partial
Multi-system	3+	1	1	1
Receipts	Native	No	No	No
FHE hybrid	Yes	No	No	No
GPU accel.	Yes	No	No	No

Table 5: Z-Chain vs existing ZK infrastructure.

proofs across chains. Formal verification in Lean 4 ensures soundness, completeness, and zero-knowledge. GPU acceleration achieves practical throughput (12,000 verifications/sec) for production deployment.

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